

Glial *betaPix* is essential for blood vessel integrity in the zebrafish brain

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This **valuable** manuscript presents findings supported by **solid** data to identify a surprising glia-exclusive function for betapix in vascular integrity and angiogenesis. The manuscript also describes the optimisation of a modified CRISPR-based Zwitch approach to generate conditional knockouts in zebrafish.

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Abstract

The formation of blood-brain barrier and vascular integrity depends on the coordinative development of different cell types in the brain. Previous studies have shown that zebrafish *bubblehead* (*bbh*) mutant, which has mutation in the *betaPix* locus, develops spontaneous intracerebral hemorrhage during early development. However, it remains unclear in which brain cells *betaPix* may function. Here, we established a highly efficient conditional knockout method in zebrafish by using homology-directed repair (HDR)-mediated knockin and knockout technology, and generated *betaPix* conditional trap (*betaPix^{ct}*) allele in zebrafish. We found that *betaPix* in glia, but neither neurons, endothelial cells, nor pericytes, was critical for glial and vascular development and integrity, thus contributing to the formation of blood-brain barrier. Single-cell transcriptome profiling revealed that microtubule aggregation signaling stathmins and pro-angiogenic transcription factors *Zfhx3/4* were down-regulated in glial and neuronal progenitors, and further genetic analysis suggested that *betaPix* acted upstream on the PAK1-Stathmin and *Zfhx3/4*-Vegfa signaling to regulate glia migration and vascular integrity. Therefore, this work reveals that glial *betaPix* plays an important role in brain vascular integrity in zebrafish embryos and possibly human cells.

Introduction

Intracerebral hemorrhage (ICH) is a life-threatening stroke type with the worst prognosis and few proven clinical treatments. Most patients who survive intracerebral hemorrhage end up with disabilities and are at risk for recurrency, cognitive decline, and systemic vascular issues, making this condition particularly significant among neurological disorders ^{1,2}. Several rodent models have been used for modeling ICH, including autologous blood injection, collagenase injection, thrombin injection, and micro-balloon inflation techniques ³. Zebrafish (*Danio rerio*) exhibit a closed circulatory system and regulatory pathways that are highly conserved among vertebrates ⁴ with relatively cheap and easy to maintain disease models. Over the past decades, varieties of zebrafish mutants that spontaneously developed brain hemorrhage have been generated in mutagenesis screens, such as *redhead* ⁵, *reddish* ^{6,7} and *bubblehead (bbh)* ^{8,9}. Elucidating mutated gene function in these zebrafish models enables us to gain insights into the genetic basis and pathophysiological mechanisms of ICH development. The *bbh* mutant was identified independently in two large-scale ethylnitrosourea-induced mutagenesis screens ^{8,9} and positional cloning revealed that p21-activated kinase (Pak)-interacting exchange factor beta (*betaPix*) gene was mutated in *bbh* mutants ¹⁰. *betaPix* contains SH3 domain that binds group I PAKs, and Dbl homology (DH) and pleckstrin homology (PH) domains that function as a RhoGEF to interact with Rac/Cdc42 small GTPases ¹¹, thus participating in multiple cellular pathways to regulate cell polarity, adhesion and migration ¹².

Various studies have reported that brain vessels are highly supported by neurons and glial cells during development and regeneration ^{13,14}. Glial-specific *betaPix* have been proposed to interact with *avβ8* integrin and the Band 4.1s in glia, which link to multiple intracellular signaling effectors such as *Wnt7a* and *Wnt7b* ¹⁵, but this hypothesis was not experimentally addressed ¹⁶. To this end, *betaPix* has been shown to interact with *avβ8* integrins and mediate focal adhesion formation and thus modulate cerebral vascular stability in *bbh* zebrafish mutant ¹⁷. Integrins are the main molecular link between cells and the extracellular matrix (ECM) which serve as scaffolds in the perivascular space. Based on a chemical suppressor screening of *bbh* hemorrhages, we have previously reported that a small molecule called miconazole downregulated the pErk-matrix metalloproteinase 9 (Mmp9) signaling to reduce ECM degradation, thus improving vascular integrity ¹⁸. However, how *betaPix* cell-type specifically contributes to vascular integrity and maturation remains unclear. Here, we generated *betaPix* conditional trap (*betaPix^{ct}*) alleles by using a CRISPR/HDR-based Zwitch method, and found that *betaPix* acted mainly in glia to regulate glial and vascular integrity via Stathmin and Zfhx3/4 signaling. Therefore, this work provides the first experimental evidence that *betaPix* acts in glia for vascular integrity development, and its functional conservation between zebrafish and human cells may further guide us to decipher the genetic basis of ICH in the future.

Results

Generating *betaPix* conditional trap (*betaPix^{ct}*) allele by an HDR-mediated knockin and knockout method

To study *betaPix* function in cell-specific manner, we utilized a donor vector with an invertible gene trap cassette via HDR (Zwitch) ^{19–21} for generating *betaPix* conditional trap (*betaPix^{ct}*) alleles. Zwitch consists of a splice acceptor conjugated to a red fluorescent protein via a 2A self-cleaving peptide, flanked by two LoxP sites and two Lox 5171 sites in reciprocal orientations. Outside this conditional trap cassette, we included a lens-specific enhanced green fluorescence protein α -crystallin:EGFP for screening knockout founders, as well as inserted both left and right

arms for targeting homologous sequences on the genome. We modified the Zwitch vector by adding a glycine-serine-glycine (GSG) spacer in front of the 2A peptide to enhance cleavage efficiency, and adding universal guide RNA (UgRNA)²² target sequences outside the left and right homologous arms for linearizing donor vector (**Figures 1A** and S1A). We chose CRISPR/Cas9 system to generate double-strand breaks (DSBs) on intron 5 of the *betaPix* locus, and chose the most efficient guide RNAs based on their efficiency to induce CRISPR indels. We used either long arms such as ~1000 bp or short arms such as 24 bp that located upstream and downstream the guide RNA sites. We then co-injected the donor vectors with targeting guide RNA, universal guide RNA, and Cas9 mRNA into one-cell-stage embryos. At 4 days post-fertilization (dpf), we selected α -crystallin-EGFP-positive embryos (F_0) for raising to adulthood, and pre-selected F_0 founders by examining inheritable EGFP reporter expression of F_1 embryos. To confirm the correct homologous recombination in the *betaPix* locus, we identified potential founders by genomic PCR to examine Zwitch insertion in the *betaPix* locus. Precise knock-in genotypes were further verified using Sanger sequencing (**Figures 1B, 1C**, S1B). Among 184 adult F_0 founders with α -crystallin-EGFP expression, we found 84 founders had EGFP-positive F_1 embryos, and 7 founders had expected PCR fragments around both 5' and 3' arms, which two founders had correct insertions in the *betaPix* locus by Sanger sequencing (Figure S1C), thus achieving ~1% efficiency on generating conditional alleles. Briefly, we established an efficient method for generating conditional knockin/knockout mutants in general, and generated a conditional gene trap line Ki(*betaPix*:Zwitch) in the *betaPix* locus in particular, which is referred to as *betaPix*^{ct} hereafter.

A previous work has shown *bubblehead* (*bbh*^{m292}) mutant with a hypomorphic mutation in *betaPix*, and *bbh*^{fn40a} mutant with reduced *betaPix* transcripts without finding mutations in the coding region and the splicing sites.¹⁰ *bbh*^{fn40a} mutants developed cerebral hemorrhages phenotype between 36 to 52 hpf (Figure S1D and S1E). To characterize the *betaPix*^{ct} allele, we injected one-cell-stage *betaPix*^{ct/ct} embryos with *Cre* mRNA to induce global inversion of Zwitch and deletion of *betaPix*. As expected, we found *Cre*-induced precise inversion in *betaPix*^{m/m} as confirmed by Sanger sequencing (**Figure 1C**) and severe brain hemorrhages developed in the time window similar to *bbh*^{fn40a} mutants (**Figures 1D** and **1E**). Consistently, qPCR analysis showed that *betaPix* decreased and *RFP* increased in heterozygous *betaPix*^{m/+} and homozygous *betaPix*^{m/m} mutants compared with *betaPix*^{ct/ct} wild-type siblings, while *alphaPix* expression was not affected (**Figure 1F**). Of note, *TagRFP* expression in *betaPix*^{m/+} (after *Cre*-mediated recombination in *betaPix*^{ct/+} embryos) reflected endogenous *betaPix* expression in the boundary of midbrains and hindbrains as well as the hindbrains (Figure S1F and S1G). Together, these results suggest that the conditional trap *betaPix*^{ct/ct} zebrafish is functional for visualizing endogenous *betaPix* expression (knockin) and loss-of-function study (conditional knockout).

***betaPix*^{m/m} mutant has brain hemorrhages, central arteries defects and abnormal glial structure that is partially rescued by Pak1 inhibitor IPA3 treatment**

The main phenotypes in *bbh* mutants consist of brain hemorrhage and hydrocephalus as early as 36 hpf, as well as poor endothelial–mesenchymal contacts and defective central arteries (CtAs) sprouting in the hindbrain.^{10,17} To demonstrate the practicability of *betaPix* conditional trap lines, we assessed vasculature and glia in the hindbrain by using light sheet fluorescence microscopy. Global *betaPix* inactivation in *betaPix*^{m/m} mutants resulted in severe cerebral hemorrhages and defective angiogenesis in hindbrain central arteries (**Figures 1D, 1E**, S2A and S2B), which is consistent with previous studies^{10,18}

It has been suggested that endothelial *betaPix* might not contribute to the occurrence of brain hemorrhage.¹⁰ During brain development, neurons and glia are important perivascular cells and they orchestrate with endothelial cells in a temporal and spatial pattern. Thus, we studied

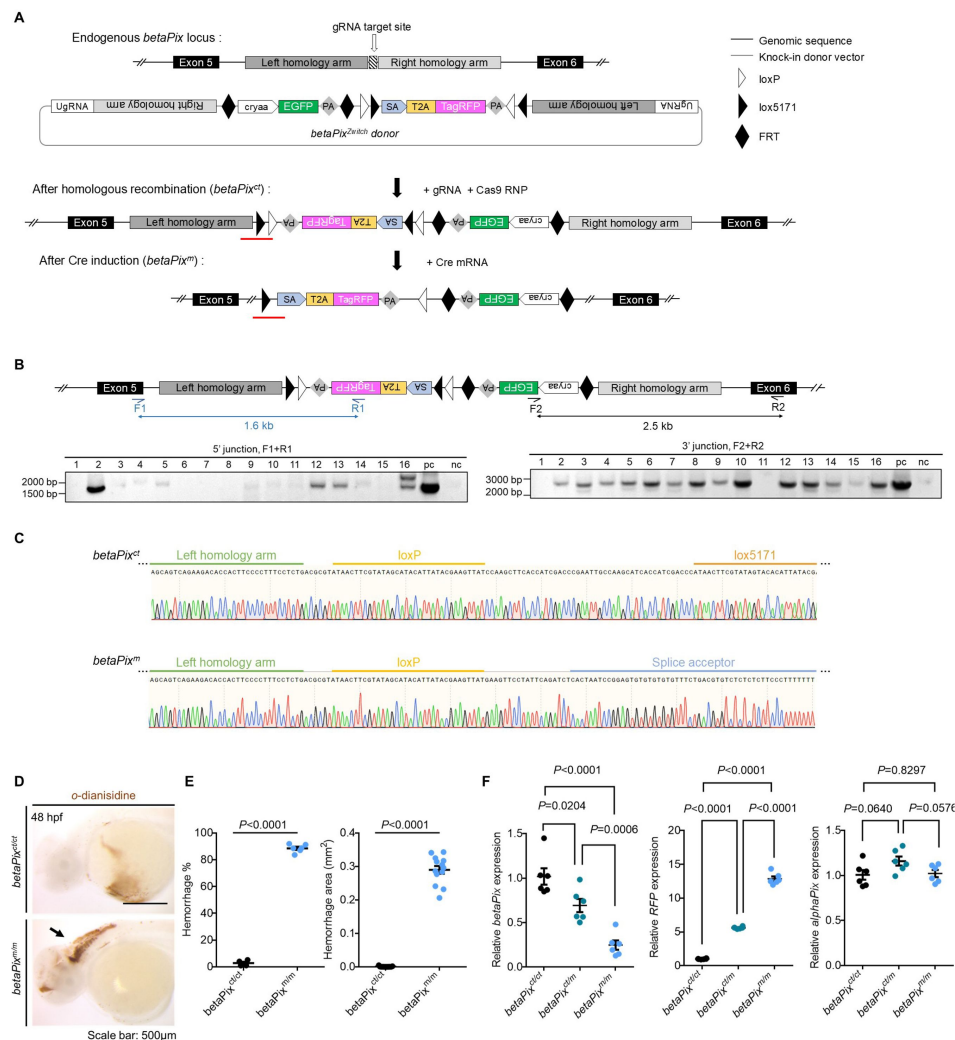


Figure 1.

Generation of *betaPix* conditional trap (*betaPix^{ct}*) allele by an HDR-mediated knock-in method.

(A) Schematic diagram illustrates the HDR-mediated *Zwisch* strategy for generating zebrafish knock-in allele at the *betaPix* locus.

(B) Genomic PCR analysis of the F₁ embryos confirming the right *Zwisch* insertions.

(C) Sanger sequencing confirming the junction of *betaPix^{ct}* (after HDR-mediated insertion) or *betaPix^m* (after Cre-mediated inversion) that are highlighted by RED lines in (A).

(D) Representative stereomicroscopy images of erythrocytes stained with o-dianisidine in *betaPix^{ct/ct}* and *betaPix^{m/m}* embryos at 48 hpf. Brain hemorrhages, indicated with arrows, in Cre mRNA-injected embryos (*betaPix^{m/m}*). Lateral views, anterior to the left.

(E) Quantification of hemorrhagic parameters in (D). Left panel showing hemorrhage percentages with each dot representing an individual experiment (n>100 each), and right panel showing hemorrhage areas with each dot representing one embryo.

(F) qRT-PCR analysis showing the expression of *betaPix*, *RFP* and *alphaPix* in *betaPix^{ct/ct}*, *betaPix^{ct/m}*, and *betaPix^{m/m}* embryos at 48 hpf. Each dot represents one embryo. cryaa, αA-crystallin; PA, polyadenylation signal; SA, splice acceptor; T2A, T2A self-cleaving peptide.

hindbrain radial glia development and found that radial glial structure had disoriented arrangement and shorten process length in *betaPix*^{m/m} embryos compared to siblings (**Figures 2A, 2B** [↗](#) and S2C). As a result, neuronal and glial precursors marker *nestin* increased while differentiated neuronal marker *pax2a* decreased in *betaPix*^{m/m} embryos, suggesting delayed neuronal development and differentiation after *betaPix* inactivation (**Figure 2C** [↗](#)). By using global *betaPix* knockout with multiple guide-RNA simultaneously ²³[↗](#), we found that CRISPR-induced *betaPix* mutant F₀ embryos had almost identical phenotypes to those in *betaPix*^{fn40a} mutants (Figure S3A-S3B), such as cerebral hemorrhages, central arteries defects, and abnormal hindbrain glia and neuronal precursor development (Figures S3C to S3G), further confirming the loss-of-function phenotypes in *betaPix*^{m/m} mutants.

Zebrafish central arteries vascularization in the hindbrains starts at 29 hpf. Tip cells start to form in the dorsal side of the primordial hindbrain channels (PHBC), sprout and migrate into the hindbrain tissue, moving towards the center around 36 hpf to establish connections with the basilar artery. A characteristic arch architecture forms at 48 hpf as boundaries dividing the hindbrain into rhombomeres²⁴[↗](#). In *betaPix* knockouts, CtA sproutings were restricted to the bottom of glial processes, thus not being able to migrate upwards to form an arched structure (**Figure 2D** [↗](#)). The transcription factor SRY box-2 (Sox2) is another marker gene for neuron and glial precursors. Sox2-positive precursor cells tightly wrapped around the central artery in control siblings, but Sox2 precursor cells had loose contacts with the central arteries, with larger perivascular distances in *betaPix* mutants (**Figure 2E** [↗](#)). These results implicate the impaired interactions between endothelial cells and neuronal/glial cells during development after *betaPix* deletion.

p21-activated kinase (Pak) is a binding partner for *betaPix*. *Pak2a* has been shown mediating downstream signaling in *bbh*^{m292}[↗](#) mutants. Group I PAK allosteric inhibitor IPA-3 covalently modifies and stabilizes the autoinhibitory N-terminal region of PAK1, PAK2 and PAK3²⁵[↗](#). As expected, IPA-3 treatment significantly decreased incidence and intensity of cerebral hemorrhages in *betaPix*^{m/m} mutant embryos (**Figure 2F** [↗](#) and **2G** [↗](#)), while IPA-3 treatment had no effect on hematopoiesis in *betaPix*^{ct/ct} control siblings. In addition, central arterial defects were partially rescued in *betaPix*^{m/m} mutant embryos after IPA-3 treatment, showing increased endothelial protruding into the hindbrain and more arch structure formation (**Figure 2H** [↗](#) and **2I** [↗](#)). IPA-3 treatment also decreased abnormal glial process arrangements, which the lengths of glial processes statistically reached the levels of the control siblings (**Figure 2J** [↗](#)). Thus, these data suggest that the *betaPix*-Pak1/2 signaling regulates brain vascular integrity during development.

Glial-specific *betaPix* knockouts recapture its global knockout phenotypes

To investigate *betaPix* function in different types of perivascular cells, we generated glial- or neuronal-specific transgenic lines using either glial fibrillary acidic protein (*gfap*)²⁶[↗](#) or *huc*²⁷[↗](#) promoters to drive *EGFP-Cre* fusion gene expression under *betaPix*^{ct/ct} background, respectively (Figure S4A). EGFP signals reported *Cre* expression while RFP signals efficiently reported *Cre*-induced gene trap cassette inversion and disruption of *betaPix* expressions (Figure S4B to S4F). Interestingly, glial-specific *betaPix* knockouts developed severe cerebral hemorrhages and abnormal central arteries vascularization, which were partially rescued by IPA-3 treatment (**Figure 3A** [↗](#) to **3C** [↗](#)), highlighting *betaPix* function in glia via Pak1/2 signaling. On the other hand, neither vascular-, neuronal-, nor mural-specific deletion of *betaPix* had evident *betaPix* mutant phenotypes (Figure S5A to S5D). These results suggest that glial *betaPix* plays crucial roles in zebrafish embryonic vascular integrity and glial development via regulating *Pak* activities.

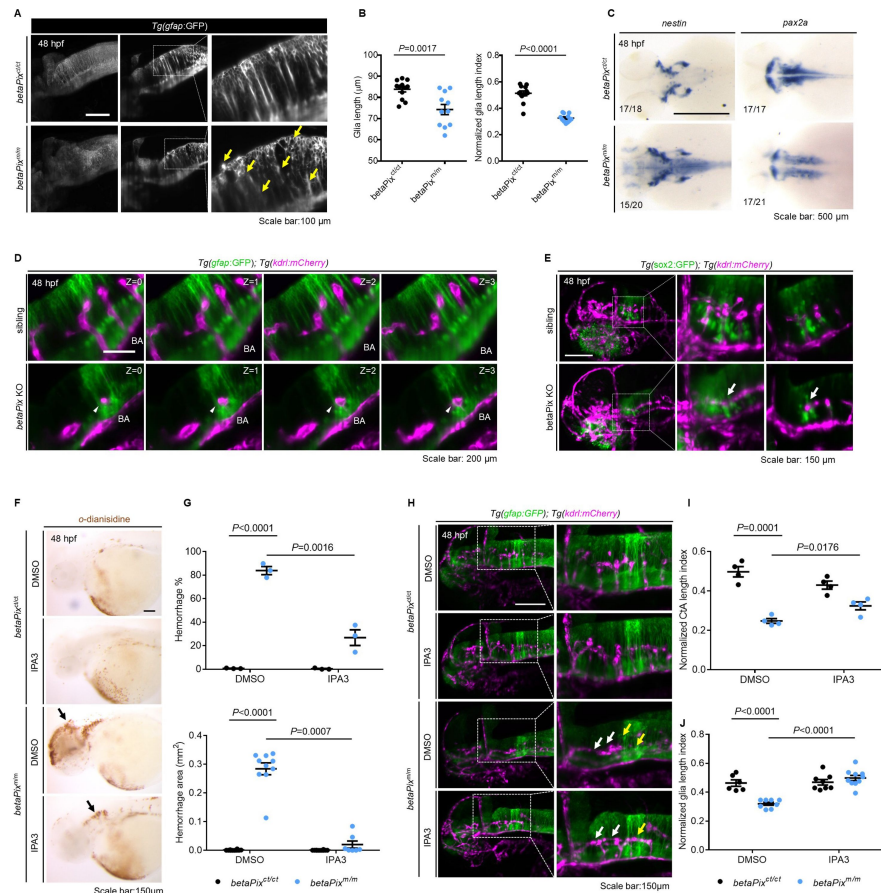


Figure 2.

***betaPix*^{m/m} mutant have brain hemorrhages, central artery defects and abnormal glial structure that was partially rescued by Pak1 inhibitor IPA3 treatment.**

(A) Left panel showing the maximum intensity projection of the glial structures in the hindbrain of *betaPix*^{ct/ct} and *betaPix*^{m/m} embryos at 48 hpf. Lateral view, anterior to left. Middle panel showing representative optical sections and right panel showing the higher magnifications of boxed area, presenting atypical glial structures with disoriented arrangements (yellow arrows) in *betaPix*^{m/m} embryos.

(B) Quantification of glial parameters in (A). Left panel showing the average glia length, and right panel showing glia length index normalized to individual head length, which each dot represents one embryo.

(C) Whole-mount RNA *in situ* hybridization revealed *nestin* and *pax2a* expression pattern in *betaPix*^{ct/ct} and *betaPix*^{m/m} embryos at 48 hpf. Dorsal view, anterior to the left.

(D) Optical sections of glial structure (green) and blood vessels (magenta) in the heads of siblings and CRISPR-mediated *betaPix* F₀ knockout embryos. Arteries in the hindbrain of *betaPix* KO mutants had developmental defects (white arrowheads), showing shorter distance between basilar artery (BA) and glial cell bodies.

(E) 3D reconstruction of the *sox2*-positive precursors (green) and vasculatures (magenta) in the heads of siblings and CRISPR-mediated *betaPix* F₀ knockout embryos at 48 hpf. Box areas are shown in higher magnifications at the middle panels, with optical sections shown in the right panels. Arrows indicate CtA with enlarged perivascular space.

(F) Representative stereomicroscopy images of *o*-dianisidine staining of *betaPix*^{ct/ct} and *betaPix*^{m/m} embryos at 48 hpf that were treated with DMSO or PAK inhibitor IPA3. Brain hemorrhages indicated with the arrows.

(G) Quantification of brain hemorrhagic parameters in (F).

(H) 3D reconstruction of the glial structure (green) and vasculature (magenta) in the heads of *betaPix*^{ct/ct} and *betaPix*^{m/m} embryos at 48 hpf treated with DMSO or IPA3. Lateral view, anterior to left. Defects in CtAs (white arrows) and glia (yellow arrows) shown in *betaPix*^{m/m} embryos treated with DMSO or IPA3.

(I) Quantification of CtA parameters in (H).

(J) Quantification of glia parameters in (H).

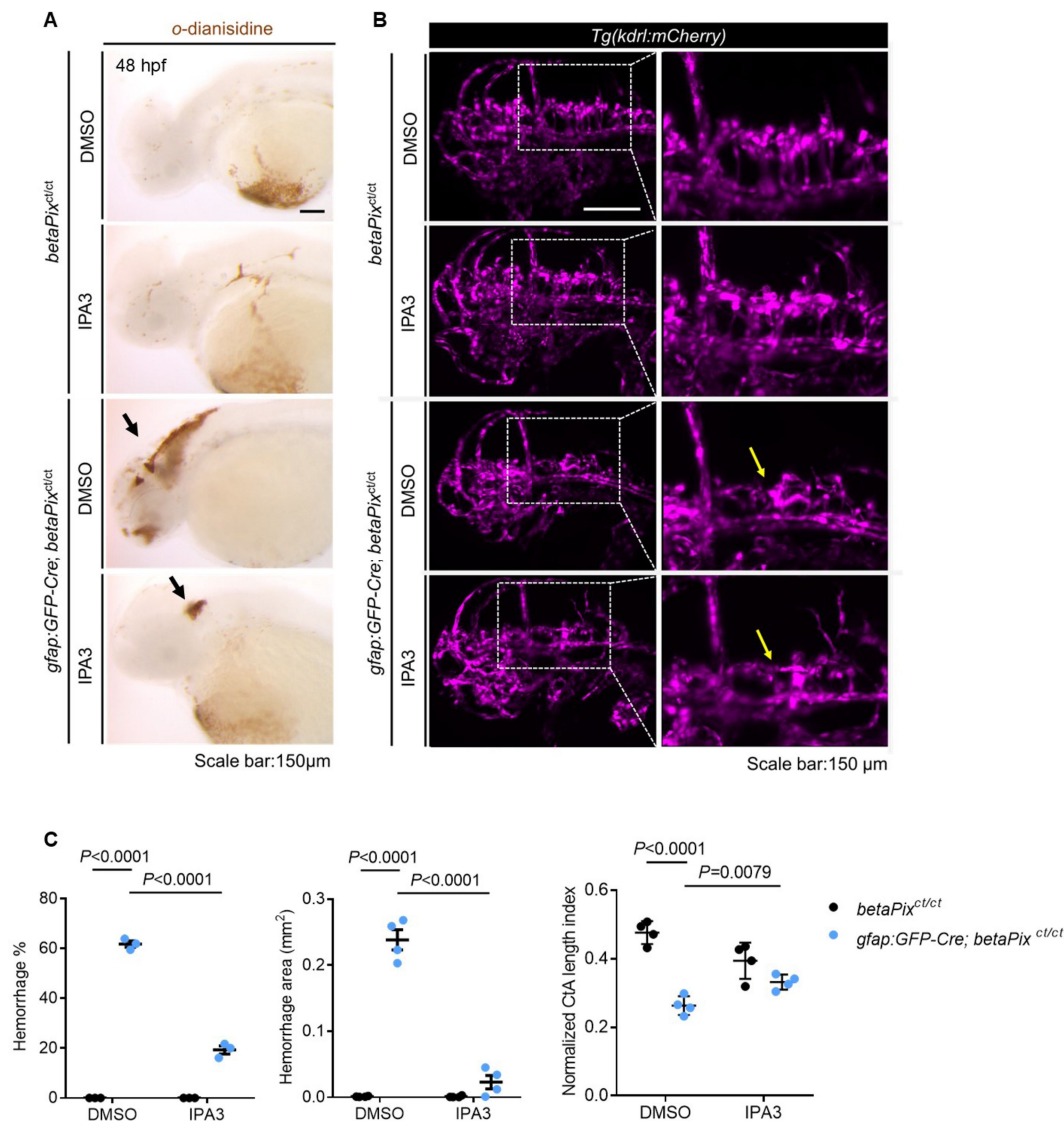


Figure 3.

Glial-specific *betaPix* knockouts recapture global *betaPix* mutant phenotypes.

(A) Representative stereomicroscopy images of erythrocytes stained with *o*-dianisidine in *betaPix^{ct/ct}* siblings and *gfap:GFP-Cre; betaPix^{ct/ct}* mutant embryos treated with DMSO or IPA3 at 48 hpf. Brain hemorrhages indicated with arrows in glial-specific *betaPix* knockouts. Lateral view with anterior to the left.

(B) 3D reconstruction of the vasculatures (magenta) in the heads at 48 hpf, showing lateral view with anterior to the left. The box areas are shown in higher magnifications of brain vasculatures at the right panels. CtA defects (yellow arrows) were evident in *gfap:GFP-Cre; betaPix^{ct/ct}* mutant embryos.

(C) Quantification of brain hemorrhages (A) and CtA parameters (B). Left panel showing hemorrhage percentages with each dot representing an individual experiment ($n > 100$ each). Middle panel showing hemorrhage areas with each dot representing one embryo. Right panel showing CtA length index normalized to individual head length, with each dot representing one embryo.

Single-cell transcriptome profiling reveals that stathmins in *gfap*-positive progenitors were affected in *betaPix* knockouts

To investigate the interplays between glial and hemorrhagic pathology caused by *betaPix* loss-of-function, we profiled the cranial tissues of CRISPR-edited zebrafish at 1 and 2 dpf using the 10X Chromium single-cell RNA sequencing (scRNA-seq) platform (**Figure 4A** [↗](#)). Multi-guide targeting was able to generate almost 100% F_0 null mutants, allowing us to select mutant embryos before brain hemorrhages from 36 hpf. Low-quality cells were excluded based on the numbers of genes detected and percentages of reads mapped to mitochondrial genes per sample. A total of 38,670 cells passed quality control and were used for subsequent analyses. Uniform manifold approximation and projection (UMAP) identified 71 cell clusters, which represented 24 zebrafish cranial cell types based on known marker gene expression profiles (**Figure 4B** [↗](#) and S6A). Enriched gene markers were compared with previously annotated gene markers in the ZFIN database and literatures^{28,29} [↗](#). By comparing the proportion of cells in each sample, we found that most neuronal clusters had increased trends, while the numbers of progenitors, endothelial cells, erythrocytes, neural crests, muscles, cartilages, retinas (photoreceptor precursor cells), olfactory bulbs, or epidermis and pharyngeal arches were reduced in *betaPix* knockout heads at 2 dpf (**Figure 4C** [↗](#) and S6B). This is consistent with the open public databases of single-cell transcriptome atlas where *betaPix* weakly expressed in a broad range of cells including glia, neurons and neuronal precursors, retinas, endothelial cells, pharynx, exocrine pancreas, olfactory cells and heart cells³⁰ [↗](#). Given that *betaPix* is critical in glia during brain vascular integrity development (**Figures 3** [↗](#) and S5), we examined the *gfap* expression among each cluster, and found relatively high *gfap* expression in clusters including progenitors, hindbrain, ventral diencephalon, ventral midbrain and floor plate (Figure S6A, arrow indicates). We next focused on the progenitor cluster, as of its enriched *gfap* expression and relatively more decreased cell numbers in *betaPix* mutants. Furthermore, this progenitor cluster exhibited high-level expression of cell proliferation and cell cycle related genes (*mki67*, *pcna*, *ccnd1*, *rrm1* and *rrm2*) as well as key glial-associated genes (*gfap*, *fabp7a*, *her4.1*, *cx43*, *id1*, *fgfbp3*, *atp1a1b* and *mdka*) (Table S1). From differentially expressed genes (DEGs) in this progenitor cluster between controls and *betaPix* knockouts at 2 dpf, enriched gene ontology (GO) terms revealed three major categories: epigenetic remodeling including chromatin silencing, negative regulation of transcription, chromatin packaging and nucleosome positioning; microtubule organizations including microtubule depolymerization, regulation of microtubule polymerization and microtubule cytoskeleton organization; and neurotransmitter secretion/transportation (**Figure 4D** [↗](#)). Of these signaling genes, we were particularly interested in microtubule genes for further studies.

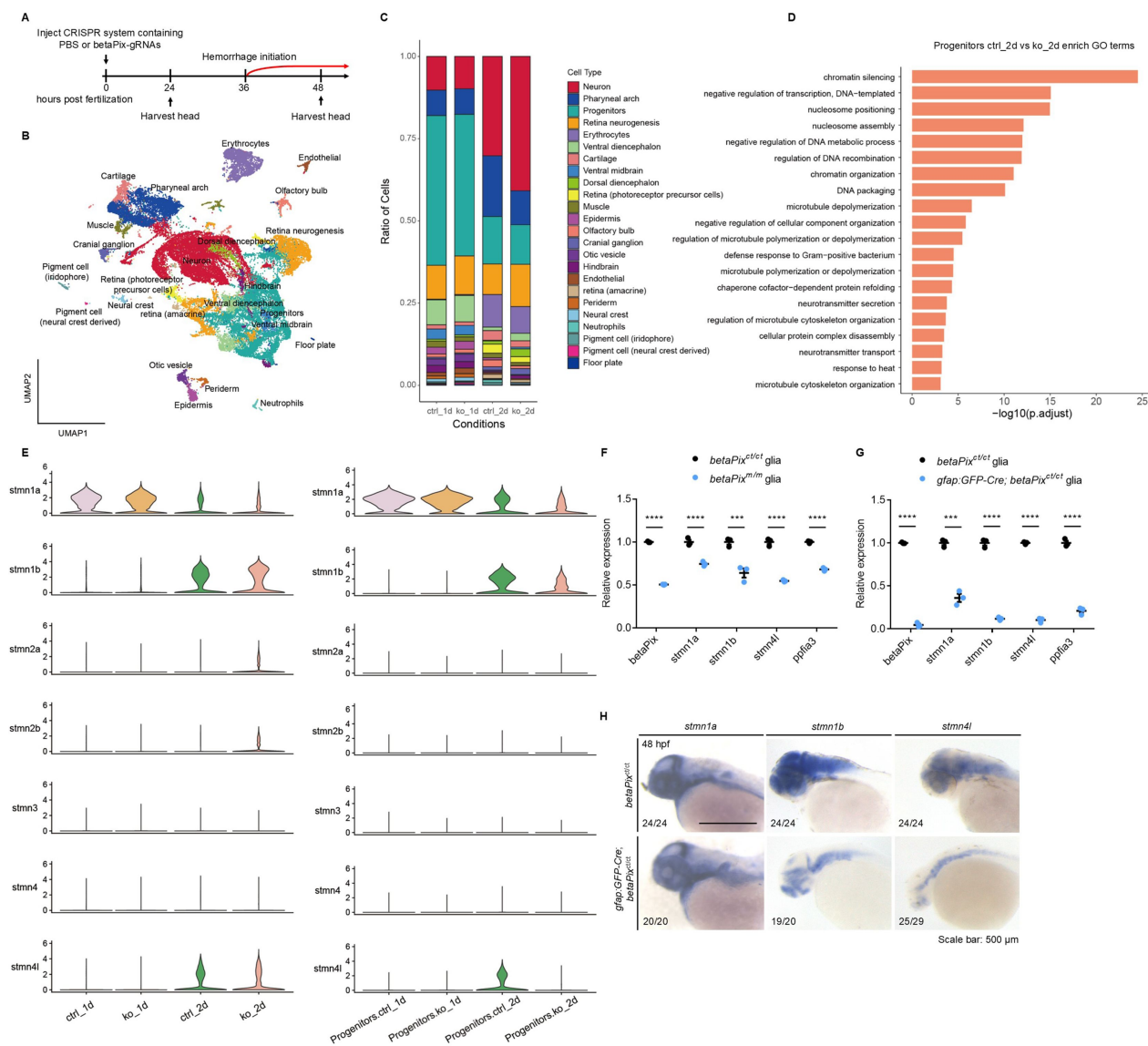


Figure 4.

Single cell transcriptome reveals that a subcluster of glial progenitor and stathmin family members are associated with *betaPix* mutation.

- (A) Experimental strategy for single cell RNA sequencing of embryonic heads from wild-type siblings and *betaPix* CRISPR mutants at 1 dpf and 2 dpf.
- (B) UMAP visualization and clustering of cells labeled by cell type. Four samples were aggregated and analyzed together.
- (C) Proportions of 24 cell clusters were differentially distributed among four sample groups. ctrl_1d, PBS-injected siblings at 1 dpf; ko_1d, *betaPix* CRISPR mutants at 1 dpf; ctrl_2d, PBS-injected siblings at 2 dpf; ko_2d, *betaPix* CRISPR mutants at 2 dpf.
- (D) Enriched GO terms for differentially expressed genes for progenitor sub-cluster comparing ko_2d to ctrl_2d groups.
- (E) Violin plots showing the expression of the stathmin family genes by all cells among four sample groups (left panel) or by progenitor sub-cluster among four sample groups (right panel).
- (F) qRT-PCR analysis showing expression of *betaPix*, *stmn1a*, *stmn1b*, *stmn4l* and *ppfia3* in glia of FACS-sorted *betaPix*^{ct/ct} siblings and *betaPix*^{m/m} mutants at 48 hpf. Each dot represents cells sorted from one embryo.
- (G) qRT-PCR analysis showing expression of *betaPix*, *stmn1a*, *stmn1b*, *stmn4l* and *ppfia3* in glia of FACS-sorted *betaPix*^{ct/ct} siblings and *gfap:GFP-Cre; betaPix*^{ct/ct} mutants at 48 hpf. Each dot represents cells sorted from one embryo.
- (H) Whole-mount RNA *in situ* hybridization revealing down-regulation of *stmn1a*, *stmn1b* and *stmn4l* in *betaPix*^{ct/ct} siblings and *gfap:GFP-Cre; betaPix*^{ct/ct} mutant embryos at 48 hpf.

Stathmin acts downstream of *betaPix* in glial migration *via* regulating tubulin polymerization

Microtubules are essential cytoskeletal elements composed of α/β -tubulin heterodimers. The regulation of microtubule polymerization affects important cellular functions such as mitosis, motor transport, and migration³¹. Based on the above scRNA-seq data (Figures 4E and S6C), we then examined if microtubule polymerization genes stathmins were affected in *betaPix* mutants. We found that stathmin1a (*stmn1a*), stathmin1b (*stmn1b*), and stathmin4l (*stmn4l*) were relatively abundant in both the whole brain and the progenitor cluster, and decreased specifically at 2dpf in *betaPix* knockout progenitor cells (Figure 4E). Subsequent qRT-PCR revealed that these stathmins expression decreased in both global *betaPix* knockouts (*betaPix*^{m/m}) and glial-specific *betaPix* knockout glia (*gfap:EGFP-Cre; betaPix*^{ct/ct}) (Figure 4F and 4G), which was further verified by RNA *in situ* hybridization analysis (Figure 4H). We next investigated whether *Stathmins* act downstream to *betaPix* signaling on regulating vascular integrity development. To this end, we found that *Pak1/2* inhibition enabled to rescue down-regulated expressions of *stmn1a*, *1b* and *4l* in either global or glial-specific *betaPix* knockout zebrafish (Figures 5A, 5B and S7A), supporting previous findings that *stathmin-1* acts downstream to *betaPix-Pak* signaling in neurite outgrowth in mice³².

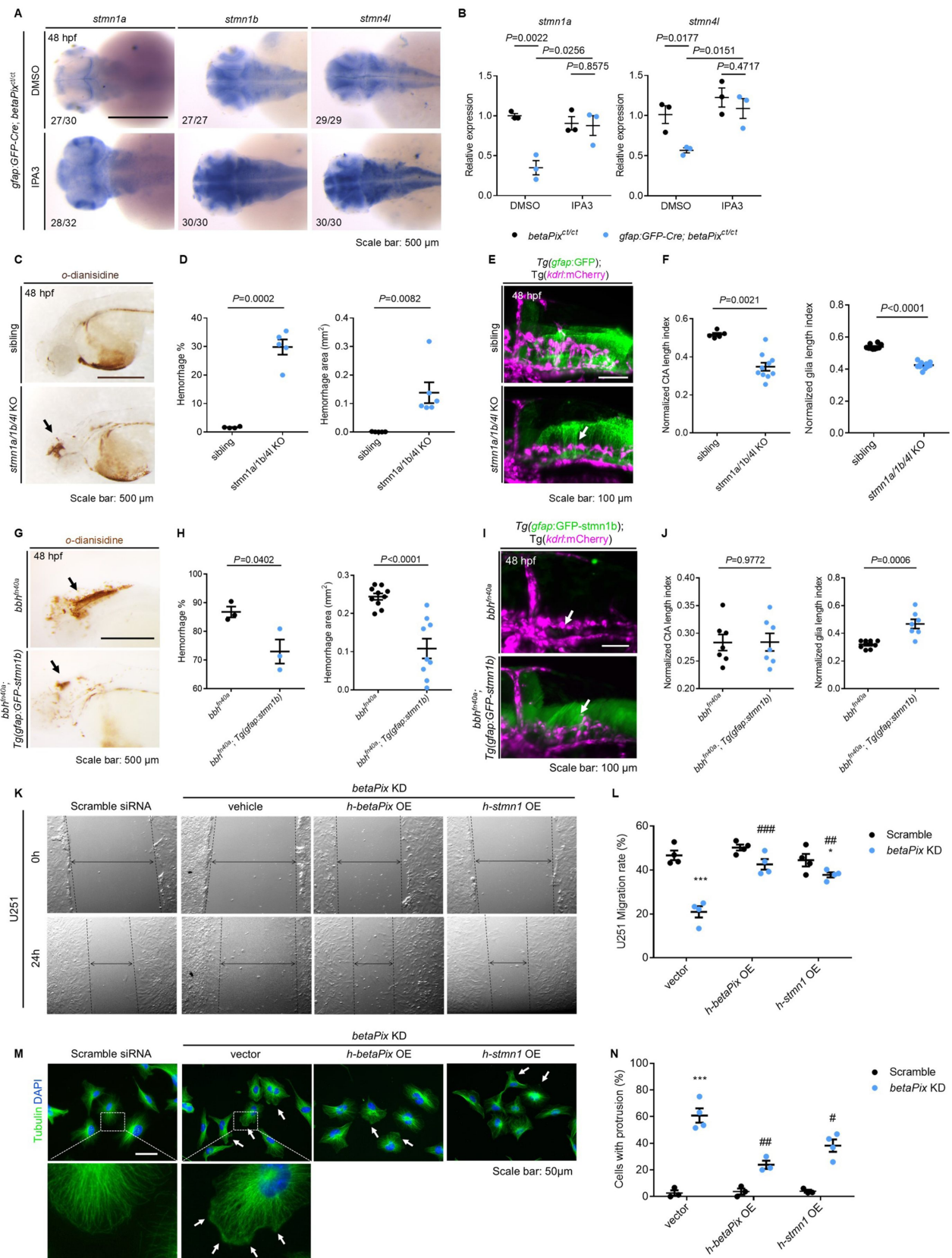


Figure 5.

Stathmin acts downstream of *betaPix* in glial migration *via* regulating tubulin polymerization.

(A) Whole-mount RNA *in situ* hybridization showing that *stmn1a*, *stmn1b* and *stmn4l* expression in *gfap:GFP-Cre; betaPix^{ct/ct}* embryos were partially rescued by Pak1 inhibitor IPA3 treatment at 48 hpf. Dorsal views, anterior to the left.

(B) qRT-PCR analysis showing that *stmn1a* and *stmn4l* expression were rescued in *gfap:GFP-Cre; betaPix^{ct/ct}* mutants by IPA3 treatment at 48 hpf. Each dot represents one embryo.

(C) Representative stereomicroscopy images of erythrocytes stained with *o*-dianisidine in siblings and *stmn1a/1b/4l* CRISPR mutants at 48 hpf. Brain hemorrhages, indicated with arrows, appeared in *stmn1a/1b/4l* mutants. Lateral views with anterior to the left.

(D) Quantification of hemorrhagic parameters in (C). Left panel showing hemorrhage percentages with each dot representing an individual experiment ($n > 100$ each), and right panel showing hemorrhage areas with each dot representing one embryo.

(E) 3D reconstruction of glial structure (green) and vasculature (magenta) in the heads at 48 hpf. Lateral view with anterior to the left. CtA defects (white arrows) appeared in *stmn1a/1b/4l* mutants.

(F) Quantification of CtA and glia parameters in (E). Length index normalized to individual head length, with each dot representing one embryo.

(G) Representative stereomicroscopy images of erythrocytes stained with *o*-dianisidine in *bbh^{fn40a}* and *bbh^{fn40a}; Tg(gfap:GFP-stmn1b)* embryos at 48 hpf. Brain hemorrhages, indicated with arrows, decreased in *bbh^{fn40a}* mutants with glia-specific overexpression of *stmn1b*, compared with *bbh^{fn40a}* mutant siblings. Lateral views with anterior to the left.

(H) Quantification of hemorrhagic parameters of (G). Left panel showing hemorrhage percentages with each dot representing an individual experiment ($n > 100$ each), and right panel showing hemorrhage area with each dot representing one embryo.

(I) 3D reconstruction of the *gfap:GFP-stmn1b* overexpression (green) and vasculature (magenta) in the heads of *bbh^{fn40a}* siblings and *bbh^{fn40a}; Tg(gfap:GFP-stmn1b)* mutants at 48 hpf. Lateral view with anterior to left. White arrows indicate CtA defects.

(J) Quantification of CtA and glia parameters in (I). Length index normalized to individual head length, with each dot representing one embryo.

(K) Representative stereomicroscopy images of U251 cells at 0 and 24 hours after wounding. U251 cells were transfected with negative control siRNA or *betaPIX* siRNA separately, in combination with pcDNA3.1 vector, *betaPIX* overexpression plasmid, and *STMN1* overexpression plasmid. The wound edges are highlighted by dashed lines, with arrow lines indicating the wound width.

(L) Quantification of wound closure in (K), showing $*P < 0.05$, $***P < 0.005$ compared to negative control siRNA with empty vector transfection. $##P < 0.01$, $###P < 0.005$ compared to *betaPix* knockdown with empty vector transfection.

(M) Representative immunofluorescence image of alpha-tubulin and DAPI signals in U251 cells. U251 cells were transfected with negative control siRNA or *betaPIX* siRNA separately, in combination with pcDNA3.1 vector, *betaPIX* overexpression plasmid, and *STMN1* overexpression plasmid. Box areas are shown in higher magnifications. Arrows

indicate protrusions at the cell periphery.

(N) Quantification of cell percentages with protrusions in (M). *** $P < 0.005$ compared to negative control siRNA with empty vector transfection. * $P < 0.05$, ** $P < 0.01$ compared to *betaPix* knockdown with empty vector transfection.

To test whether *Stathmin* family genes are important for vascular stability, we utilized CRISPR-induced F_0 knockout system simultaneously targeting the three stathmin genes. While singular stathmin gene knockout led to no evident phenotypes (data not shown), we found that simultaneously inactivating *stmn1a*, *stmn1b* and *stmn4l* caused brain hemorrhages and hydrocephalus in the brain, similar to that from *betaPix*^{fn40a} and *betaPix*^{m/m} mutants (Figures 5C, 5D, S7B to S7E). By labeling both glia and vasculature with Tg(*gfap*:GFP; *kdr*:mCherry) transgenic embryos, we found that triple stathmin gene knockouts had impaired CtAs, abnormal glia and neuronal development in the hindbrain region (Figure 5E, 5F, S7F to S7H). Furthermore, overexpressing *stmn1b* by the *gfap* promoter partially rescued brain hemorrhages and delayed neuronal development in *bbh*^{fn40a} mutants (Figures 5G, 5H, S7I and S7J), of which glial processes elongation also improved, but CtAs defects failed to restore (Figures 5I, 5J and S7K). Thus, these data support the critical role of stathmins in glia.

To examine the direct role of *betaPix-stathmin* in glia, we turned to use human glioblastoma cell lines and manipulated the *betaPix* levels by siRNA knockdown or transgenic overexpression. We found that *betaPix* knockdown attenuated glial migration (Figures 5K, 5L, S7L to S7N), which is consistent with previous studies on *betaPix* function in the endoderm³³, epithelial cells³⁴, neurons³⁵ and multiple carcinoma cell lines including glioblastoma^{36–38}. Either *betaPix* or *Stmn1* overexpression led to a partial restoration of the impaired glial migration, suggesting *Stathmins* as downstream effectors of *betaPix*. Furthermore, immunofluorescence analysis revealed that *betaPix* knockdown led to abnormal accumulation of microtubules at the cell periphery, whereas ectopic expression of *betaPix* and *Stathmin* partially rescued these microtubule defects (Figures 5M and 5N). Together, these results suggest that *betaPix* depletion disrupts microtubule dynamics and leads to impaired glial development acting upstream to the *Pak-Stathmin* signaling, subsequently disrupting vascular integrity and resulting in brain hemorrhages. While the brain hemorrhages were only partially rescued and abnormal CtAs defects failed to restore, indicating that additional pathways may act downstream of *betaPix*.

Zfhx3/4 acts downstream of *betaPix* in regulating vascular integrity development

Previous studies have identified multiple signaling pathways important for zebrafish hindbrain angiogenic sprouting, such as VEGF, Notch, Wnt, Integrin $\beta 1$, angiopoietin/TIE, and insulin-like growth factor signaling^{39–42}. To investigate whether angiogenic signal is disrupted by *betaPix* depletion, we examined several angiogenic gene expression pattern by performing RNA *in situ* hybridization of embryos at 36 hpf. The expression level of *Vegfaa* in sprouting CtAs decreased in *bbh*^{fn40a} mutants (Figure 6A). Consistently, *betaPix* knockdown significantly reduced *VEGFA* expression in cultured glial cells, which was rescued by ectopic expression of *betaPix* (Figure 6B; right panel). To explore how *betaPix* regulate *Vegf* signaling in glia, we also found a significant enrichment of epigenetic and transcriptional regulation in progenitor cluster indicated by the GO analysis of single-cell sequencing data (Figure 4D). Among these candidate genes, transcription factor Zinc Finger Homeobox 3 (ZFHX3) is known to mediate hypoxia-induced angiogenesis in hepatocellular carcinoma via transcriptional activation of *VEGFA*⁴³. Furthermore, ZFHX3 has been associated with stroke in multiple genome-wide association studies^{44–46}. Interestingly, we found that transcription factors *Zfhx3* and *Zfhx4* significantly downregulated after *betaPix* inactivation in cultured glial cells (Figure 6B), CRISPR-induced *betaPix* mutants at 2 dpf (Figure S8A), and glia-specific *betaPix* mutants (Figure S8B), thus suggesting that *betaPix* acts upstream to *Zfhx3/4-VEGFA* signaling in regulating angiogenesis.

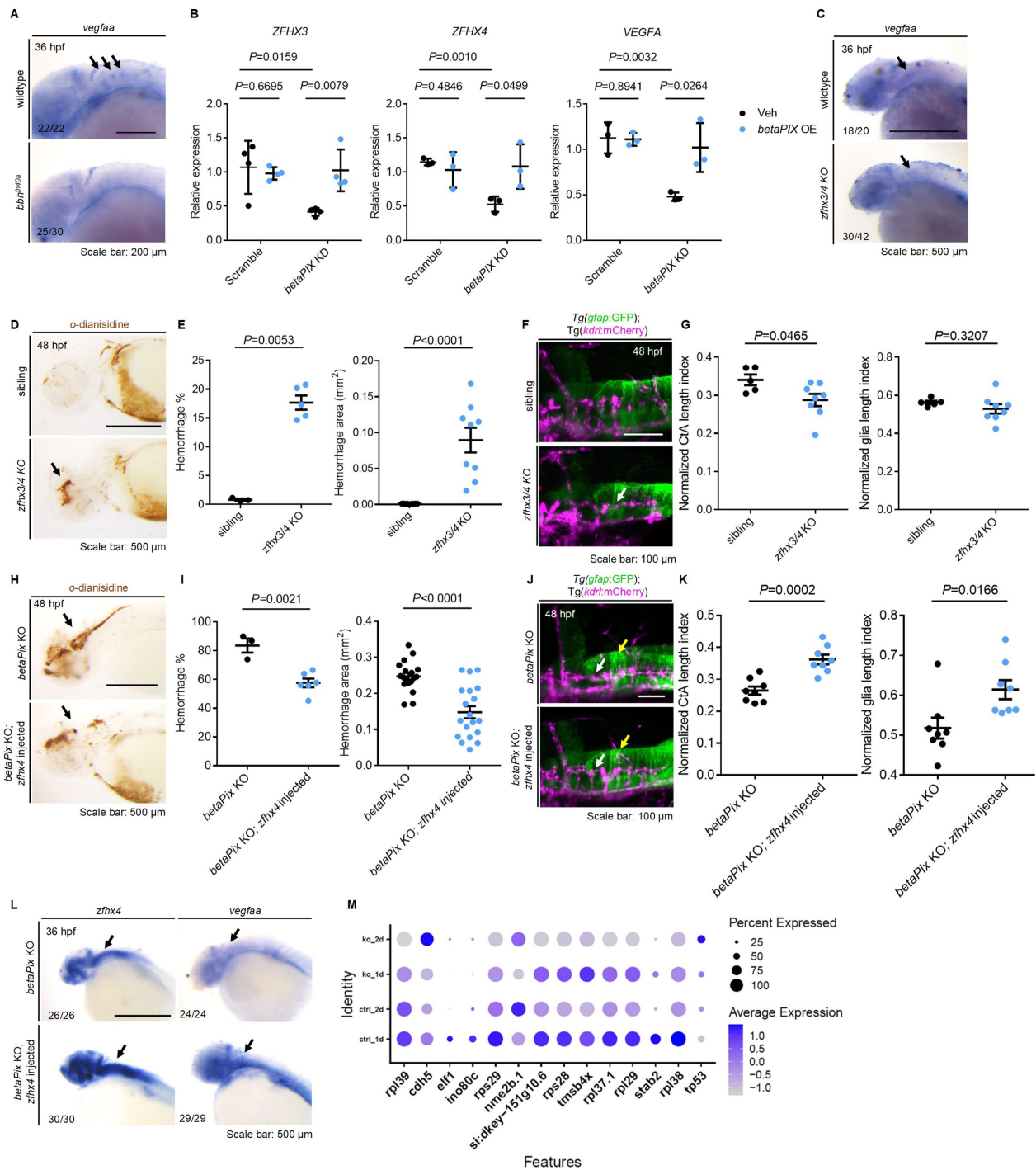


Figure 6.

***Zfhx3/4* acts downstream of *betaPix* to regulate vascular integrity development.**

(A) Whole-mount RNA *in situ* hybridization revealing that *vegfaa* decreased in *bbh^{fn40a}* mutants compared with siblings at 36 hpf. Lateral views with anterior to the left. Arrows indicate *vegfaa* expression in the CtAs.

(B) qRT-PCR analysis revealing that *ZFHx3*, *ZFHx4* and *VEGFA* decreased in U251 cells transfected with *betaPIX* siRNA, which were rescued by *betaPIX* overexpression.

(C) Whole-mount RNA *in situ* hybridization showing that *vegfaa* decreased in CRISPR-mediated *zfhx3/4* F₀ knockout embryos at 36 hpf. Lateral views with anterior to the left. Arrows indicate *vegfaa* expression in the CtAs.

(D) Representative stereomicroscopy images of erythrocytes stained with o-dianisidine in siblings and CRISPR-mediated *zfhx3/4* F₀ knockout embryos at 48 hpf. Arrows indicated brain hemorrhages in the knockout brains. Lateral views with anterior to the left.

(E) Quantification of hemorrhagic parameters in (D). Left panel showing hemorrhage percentages with each dot representing an individual experiment (n>100 each), and right panel showing hemorrhage areas with each dot representing one embryo.

(F) 3D reconstruction of the glial structure (green) and vasculature (magenta) in the hindbrains of siblings and CRISPR-mediated *zfhx3/4* F₀ knockouts at 48 hpf. Lateral view with anterior to the left. White arrows indicate CtA defects in *zfhx3/4* knockout embryos.

(G) Quantification of CtA and glia length parameters of (F). Length index normalized to individual head length, with each dot representing one embryo.

(H) Representative stereomicroscopy images of erythrocytes stained with o-dianisidine in CRISPR-mediated *betaPix* F₀ knockout embryos with or without *zfhx4* mRNA injection at 48 hpf. Arrows indicate brain hemorrhages. Lateral views with anterior to the left.

(I) Quantification of hemorrhagic parameters in (H). Left panel showing hemorrhage percentages with each dot representing an individual experiment (n>100 each), and right panel showing hemorrhage area with each dot representing one embryo.

(J) 3D reconstruction of the glial structure (green) and vasculature (magenta) in CRISPR-mediated *betaPix* F₀ knockout embryos with or without *zfhx4* mRNA injection at 48 hpf. White arrows indicate CtAs and yellow arrows indicate glia.

(K) Quantification of CtA and glia length parameters in (J). Length index normalized to individual head length, with each dot representing one embryo.

(L) Whole-mount RNA *in situ* hybridization revealed that *zfhx4* and *vegfaa* decreased in CRISPR-mediated *betaPix* F₀ knockout embryos at 36 hpf, which were rescued by *zfhx4* mRNA injection. Lateral views with anterior to the left. Arrows indicate hindbrain regions.

(M) Dot plots of several angiogenesis-associated genes expression in endothelial cell cluster. Dot size indicates the percentage of cells with gene expression, and dot color represents the average gene expression level.

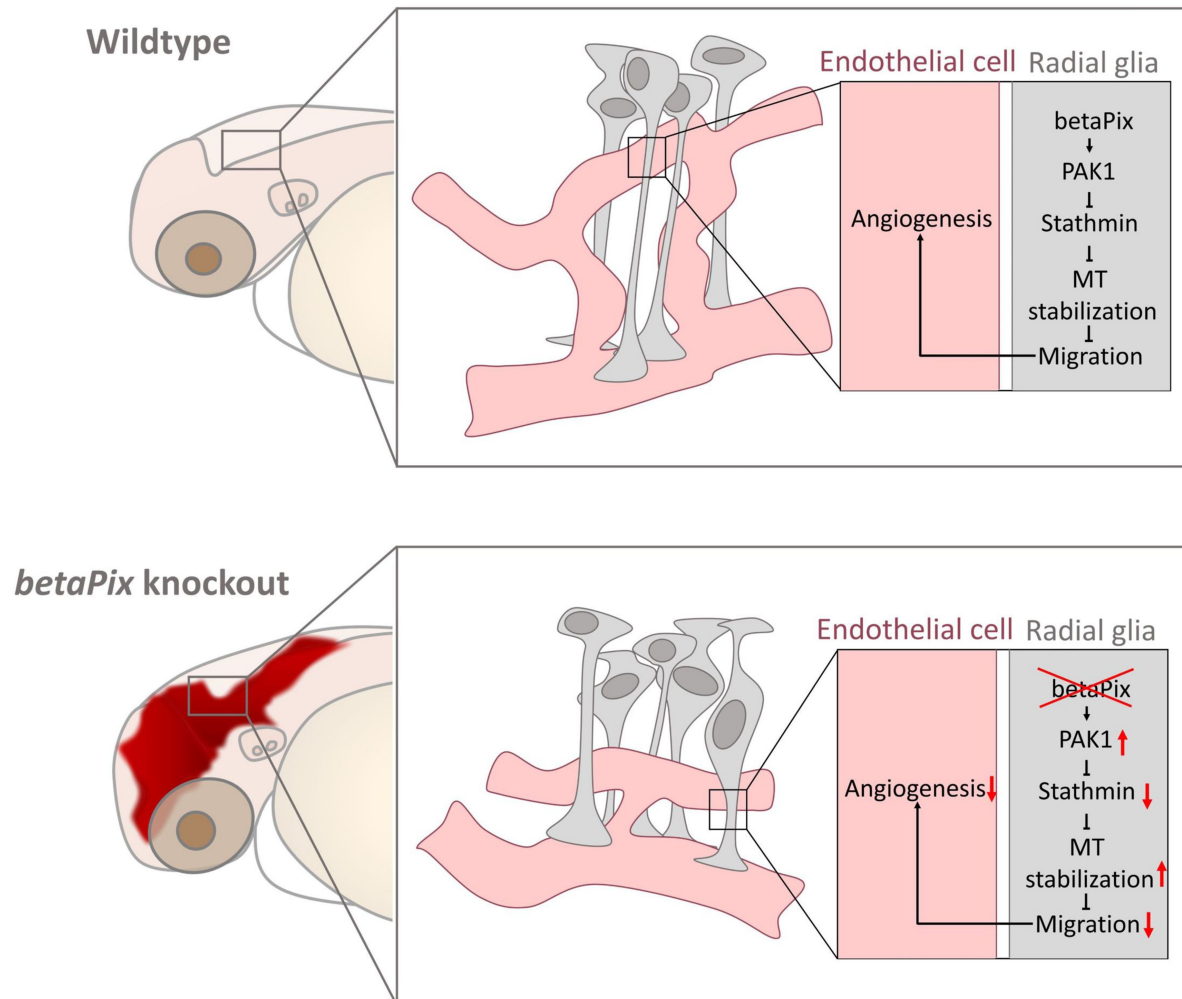


Figure 7.

Working Model on the function of glial *betaPix* in zebrafish vascular integrity development of the hindbrain.

betaPix is enriched in glia and regulates the PAK1-Stathmin axis on microtubulin stabilization, thus fine-tuning glial cell migration and their interactions with vascular endothelial cells; and in parallel, *betaPix* may also regulate Zfhx3/4-Vegfa signaling in glia, which then modulates angiogenesis during cerebral vessel development and maturation. Deletion of *betaPix* affects glial cell migration and interaction with cerebral endothelial cells. MT, Microtubule.

To determine whether *Zfhx3/4* is critical for vascular integrity by interacting with *betaPix* in zebrafish, we depleted *Zfhx3* and *Zfhx4* simultaneously via CRISPR-mediated F₀ knockouts (Figure S8C to S8E). We found that *zfhx3/4* knockouts decreased *Vegfaa* expression at sprouting CtAs at 36 hpf (Figure 6C). *Zfhx3/4* knockouts at 48 hpf had under-developed CtAs and reduced penetration of endothelial cells into the hindbrain (Figure 6F and 6G), developed cerebral hemorrhages during 36 to 52 hpf, which are comparable with *betaPix* mutant phenotypes (Figure 6D–6G). Despite up-regulated expression level of *nestin*, glial arrangement and processes elongation remained statistically unaffected after *Zfhx3/4* inactivation (Figures 6F, 6G and S8F). We next investigated whether increasing expression level of *Vegfaa* in *betaPix* mutants was able to rescue impaired vascular integrity. By using *Zfhx4* mRNA microinjection into one-cell stage embryos in either *bbh^{fn40a}* mutants or CRISPR-induced *betaPix* knockouts, we found that ectopic expression of *Zfhx4* increased *Vegfaa* expression level (Figure 6L), thus leading to a drastic reduction of both percentages and volumes of brain hemorrhage shown in these mutants (Figures 6H, 6I, S8G to S8I), as well as improved CtAs and glial development (Figures 6J and 6K). In addition, scRNA-seq data showed that CRISPR-induced *betaPix* knockouts had down-regulated angiogenic gene expression in a cluster of endothelial cells (Figure 6M). Together, these data support the critical role of *Zfhx3/4* in vascular integrity and glial development probably acting upstream to VEGFA and downstream to *betaPix*.

Discussion

Conditional knockout technology in mice has been widely used to modify and determine targeted gene function in a cell-specific manner. Spatio-temporal specific modification is beneficial for studying embryonic lethality genes or investigating the function of targeted genes in specific cell populations. Despite being broadly utilized in mice, conditional knockouts in other animal models are limited due to technical difficulties. To this end, multiple research groups have designed different strategies for establishing floxed-mutant alleles, gene traps, and inducible Cas9 in zebrafish⁴⁷. In particular, the Zwitch gene trap-based approach has been established successfully in zebrafish, enabling to simultaneously produce knockin reporter driven by the endogenous gene promoter and disrupting endogenous gene function^{19,21}. We adapted this strategy with the aid of CRISPR/Cas9 technology, and added a GSG spacer to enhance cleavage efficiency and universal guide RNA target sites on both homologous arms for producing linearized targeting vector. Moreover, we also found that utilizing either short or long homologous arms enables to achieve precise targeted integrations with this donor vector. Our short arms-based method can be broadly adapted because of less restrictions on arm sequences and easily engineering the donor vector. From the *betaPix* locus and other five unpublished loci that we already generated Zwitch alleles with ~1% correct insertions based on homologous recombination, we achieved Cre-induced, cell-specific deletion and functional analysis of targeted genes in both embryos and adult hearts. Therefore, our CRISPR-based Zwitch method can be widely applied for generating conditional mutant zebrafish in particular and possibly extended for other animal mutants in general.

Development of the vasculature in the central nervous system can be classified into three phases: vasculogenesis, angiogenesis and blood-brain barrier formation. The hallmark of CNS vasculature maturation is acquisition of BBB properties, establishing a stable environment crucial for brain homeostasis in response to extrinsic factors and physiological changes. The core features of the BBB include specialized tight junctions, highly selective transporters and limited immune cell trafficking^{48,49}. In zebrafish, BBB can be functionally characterized from 3 dpf^{50,51}. While *bubblehead* mutants acquire vessel rupture phenotype early before BBB maturation, presenting a good hemorrhagic model for studying early BBB development. Multiple types of cells contribute to diverse aspects of BBB development and maintenance, leading to proper vasculature function across developmental timeline. Pericytes are perivascular cells embedded within the same basement membrane with endothelial cells⁵². Pericyte-deficient mice with disrupting

PDGF-B signaling displayed BBB leakiness, reflecting altered pericyte coverage and endothelial gene expressions ^{53–55}. Pericyte-secreted signals promoting the endothelial survival, proliferation and sprout formation via multiple pathways such as transforming growth factor- β (TGF- β) signaling ^{56–59}, VEGF-VEGFR signaling ^{60–61}, Angiopoietin-Tie2 signaling ^{62–64}, Sphingosine-1-phosphate (S1P) signaling ⁶⁵ and Notch signaling ^{66–67}. Apart from endothelial cells, pericytes also regulate development of other members of the neurovascular unit. Pericytes regulate astrocyte aquaporin-4 polarization and guide end-foot processes toward endothelial tubes ^{53–68}. In addition, pericytes secrete ECM molecules supporting the basement membrane and regulation of the angioarchitecture ^{69–72}.

Endothelial cells form the continuous inner layer of blood vessels with junctional protein and selective transporters, which are the main subject of permeability regulation in BBB. It is well accepted that endothelial cells do not show a predetermined role and that the brain environment is sufficient to induce endothelial barrier ⁷³. Accumulative evidences also point to critical role of endothelial signals in regulating several aspects of BBB development. In particular, endothelial cell-derived PDGF-B directly binds to surface receptor PDGFR β on pericytes that are critical for pericyte recruitment and function ^{74–75}. Although their barrier properties are non-assessed, angiocrine signals such as leukaemia inhibitory factor (LIF) and TGF- β induce both astrocyte ⁷⁶ and oligodendrocyte differentiation ⁷⁷, respectively. Wnt7a/b ligands secreted by neural progenitor cells bind to Frizzled receptors on endothelial cells, activating canonical Wnt signaling and downstream genes. WNT ligands knockouts lead to angiogenic defects and BBB leakages specifically in the CNS ⁷⁸. Endothelial GPR124, as a coactivator of canonical Wnt signaling, has important function in CNS-specific angiogenesis and BBB establishment ^{79–81}. Other signaling ligands such as angiopoietins and Ephrin family ligands orchestrate with Tie or EphB receptors on endothelial cells, respectively ⁸². Briefly, complex interactions between BBB cell types orchestrate proper angiogenesis and CNS development in a spatio-temporal manner.

Compared with pericytes and endothelial cell signaling, we have less knowledge on glia signaling in the BBB development and maturation. Astrocytes ensheath capillaries through polarized end feet that are enriched with aquaporin-4 (Aqp4) proteins, colocalized with inwardly rectified K⁺ channels ^{83–84}. Functional coupling of ion and water fluxes plays a critical role in regulating local osmotic equilibrium. During development, radial glia is a neuroepithelial origin with heterogeneous populations that are able to generate neurons, astrocytes, and oligodendrocytes ^{85–88}. In zebrafish, radial glia has long been considered serving an astrocytic role. Radial glia in zebrafish enriches with several key molecular markers for astrocytes and tight junctions while Aqp4⁺ radial glial processes rarely contact the vasculature ⁸⁹. A more recent article has reported that zebrafish *Glast*-expressing radial glia transform into astrocyte-like cells, displaying dense cellular processes, tiling behavior and proximity to synapses ⁹⁰. Whether end feet of these astrocytes enwrap capillary blood vessels and resemble mammalian astrocytes warrant future investigations. In agreement with mammalian astrocytes, two independent groups have shown that ablation of pan-glia results in progressive brain hemorrhage ^{91–92} or deficiency of spinal cord arteries development in zebrafish ⁹³, demonstrating important role of zebrafish glia in BBB formation. Multiple glia/astrocyte-derived signals have been shown to contribute to endothelial barrier properties development. Astrocyte-expressed sonic hedgehog (Shh) bind to hedgehog (Hh) receptors on endothelial cells and contribute to the BBB functions by promoting junctional protein expression and the quiescence of the immune system ^{94–97}. Before BBB formation, notochord-derived Shh activity promote arterial cell fate on developing endothelial cells ⁹⁸. Shh stimulates the production of angiogenic cytokines, angiopoietins and interleukins, via downstream transcription factor *Gli* or non-canonical pathways crosstalks with iNOS/Netrin-1/PKC, RhoA/Rock, ERK/MAPK, PI3K/Akt, Wnt/ β -catenin, and Notch signaling pathways ⁹⁹. VEGF is the main pro-angiogenic growth factor produced by various cell types that controls multiple vascular development steps, such as proliferation, migration, permeability and survival via interacting with VEGF receptors on endothelial cells. Inactivation of VEGFA in astrocytes leads to BBB disruption in multiple sclerosis models ^{100–101}. In early development of avascular

neural tube, VEGF derived from neuroectoderm serve as a central role in inducing tip cells sprouting from the perineural vascular plexus (PVNP) and formation of vasculature lumens from stem cells^{102,103}. Ablating neuroglia reduces *vegfab* signals and leads to ectopic intersegmental vessels sprouting and vertebral arteries in zebrafish⁹³. VEGF signaling orchestrates elaborately with other signaling pathways such as delta-like 4 (Dll4)/Notch signaling¹⁰⁴ and activates downstream components including the Ras/Raf/MEK, PI3K/AKT, and p38/MAPK/HSP27 pathways¹⁰⁵. Our data suggest glial *betaPix* as an important factor in vascular integrity development. Together with previous studies, *betaPix-Pak1-stathmin* signaling regulates glial growth and differentiation³² and *betaPix* regulates *Vegfaa* secretion via downstream transcription factors *Zfhx3/4*. Thus, this work presents a novel glial *betaPix* signaling in regulating zebrafish BBB development.

Basement membrane is a non-cellular component consisting of several extracellular matrix (ECM) proteins, which laminins are the most abundant ones, while collagen IV, fibronectin, and perlecan serve as hub for structural supports, anchoring and intercellular communication. Components of basement membrane are synthesized in surrounding cells with diverse expression patterns¹⁰⁶. For example, mice with deletion of Collagen IV in either brain microvascular endothelial cells (BMEC)- or pericytes lead to fully penetrant intracerebral haemorrhage, while COL4A1 deletion in astrocytes results in mild haemorrhage phenotypes⁶⁹. Differential cell-specific expression pattern of laminin isoforms contribute differently to BBB formation¹⁰⁷. As astrocytic laminins mediate pericyte differentiation and regulate capillary permeability¹⁰⁸, intercellular interactions with the ECM occur by cell surface integrin receptors. Integrin $\beta 1$ on endothelial cells are critical during angiogenesis¹⁰⁹. On the other hand, integrin $\alpha v\beta 8$ has glial-specific enrichment which mediate focal adhesion to modulate vascular integrity^{17,110,111}. Notably, degradation of ECM and focal adhesions at the perivascular space is responsible for the hemorrhagic phenotypes of *bbh* mutants^{17,18}. It would be of great interest to determine whether glial *betaPix* account for ECM breakdown in *bbh* mutants in future studies.

Materials and Methods

Fish maintenance

Zebrafish were raised and handled in accordance with the guidelines of the Peking University Animal Care and Use Committee accredited by the AAALAC. The *bbh*^{h40a} mutant was isolated from a large-scale mutagenesis screen of the zebrafish genome at Massachusetts General Hospital, Boston⁹. Tg(*acta2:GFP-Cre*) was purchased from Xinjia Pharmaceutical Technology Co., Ltd (Nanjing, China); Tg(*neurod:EGFP*) were kindly provided by Dr. Bo Zhang (Peking University, China); Tg(*huC:GFP*) were kindly provided by Dr. Liangyi Chen (Peking University, China); Tg(*kdr1:GFP*)¹¹² and Tg(*kdr1:mCherry*)¹¹³ were described previously.

Cell culture and transfection

U251 and A172 human glioblastoma cell lines, kindly provided by Dr. Jian Chen at Chinese Institute for Brain Research, Beijing, were cultured in high glucose DMEM medium (Hyclone) supplemented with 10% fetal bovine serum and 1% penicillin/streptomycin, at 37°C in 5% CO₂ humidified incubator. The *betaPix* siRNAs and *STMN1* siRNAs were purchased from GenePharma (Shanghai, China). The sequences of the siRNAs are listed in Table S2. Transfection was conducted with Lipofectamine 3000 (Invitrogen) at a final siRNA concentration of 30 nM.

mRNA/gRNA synthesis and microinjections

mRNA and gRNA were synthesized as described previously¹¹⁴. In brief, linearized pT3TS-nls-zCas9-nls¹¹⁵, pXT7-*Cre*, and pXT7-*Zfhx4* plasmid DNA were purified as templates using TIANquick Mini Purification Kit (TIANGEN, DP203, China). Next, *in vitro* transcription reactions

were performed using the mMACHINE T3 (for pT3TS-nls-zCas9-nls) or T7 (for pXT7-*Cre* and pXT7-*Zfhx4*) kits (Life Technologies) according to manufacturer's instructions. Templates for gRNA were generated by complementary annealing and elongation of two oligos. Forward oligo contained a T7 promoter and gRNA target sequence, and reverse oligo contained the universal gRNA scaffold ²³. The resulting double-stranded DNA served as the templates for *in vitro* transcription using HiScribe® T7 High Yield RNA Synthesis Kit (NEB). The sequences of the gRNAs are listed in Table S3. For global *betaPix* inactivation, *Cre* mRNA with a working concentration of 200 ng/μL were injected into the *betaPix*^{ct/ct} embryos at one-cell stage. For CRISPR-mediated knockouts, 300 ng/μL Cas9 mRNA and 20-50 ng/μL gRNAs were injected into zebrafish embryos at one-cell stage. For *Zfhx4* rescue experiments, *Zfhx4* mRNA with a working concentration of 400 ng/μL were injected into the *bbh*^{fn40a} mutants or in combination with CRISPR-mediated *betaPix* knockout system at one-cell stage. Injected embryos were scored for mutant phenotypes at 36 or 48 hpf.

Construction of *betaPix* conditional knockout zebrafish

The pZwitch plasmid clone was kindly provided by Dr. Kazu Kikuchi (National Cerebral and Cardiovascular Center Research Institute, Suita, Japan). GSG spacer was added to pZwitch+3 between P2A and TagRFP coding sequences by overlapping primer pairs. Next, a highly efficient gRNA was found from the fifth intron of *betaPix*. Primer pairs were designed to amplify the left and right homologous arms from the gRNA site for 1000 bp and 24 bp with following modifications: 1) The 5' end of the forward primer of the left homologous arm were added with an *Nhe1* site and a universal gRNA site which provided an *in vivo* linearization cleavage site; a *Mlu1* site to the 5' end of the reverse primer of the left arm. 2) Added *EcoR1* site to the 5' end of the forward primer of the right homologous arm; added *Xho1* site and universal gRNA site to the 5' end of the reverse primer of the right homologous arm. Then we cloned the inverted homologous arms into polyclonal sites of the modified pZwitch+3, and purified plasmid DNA with EndoFree Mini Plasmid Kit (Tiangen, DP118). *In vivo* knock-in system was composed of 300 ng/μL Cas9 mRNA, 50 ng/μL *betaPix*-intron5 gRNA, 50 ng/μL Universal gRNA, and 20 ng/μL donor vector. Microinjected *in vivo* knock-in system into wildtype embryos at one-cell stage. Fish founders were screened for α-crystallin reporter expression, raised to adulthood, and re-screened for germline transmission and precise knock-in by Sanger sequencing.

Construction of transgenic reporters

The glial reporter plasmid *gfap*-EGFP ²⁶ was kindly provided by Dr Jiulin Du (Shanghai Institute for Biological Sciences, Shanghai, China). The *gfap* promoter was used to establish Tg(*gfap*:GFP-*Cre*) plasmid. The *stmn1b* coding sequences were amplified from wild-type zebrafish cDNA library, which were used to establish Tg(*gfap*:GFP-*stmn1b*) plasmid. Transgenic reporters were generated by using *Tol2*-based transgenesis ¹¹⁶. In brief, 100 ng/μL *Tol2* transposase mRNA in combination with 20 ng/μL donor plasmid DNA were co-injected into the embryos at one-cell stage. Transgenic founders were screened for specific transgenic GFP expression, raised to adulthood, and re-screened for germline transmission.

Inhibitor treatment

IPA-3 (Proteintech, CM05727) were dissolved in DMSO to form a 10 mM stock, and then diluted with E3 medium to 3 μM in 6-well plates. Zebrafish embryos were treated from 24 to 48 hpf, and then washed with E3 medium for phenotype analysis.

o-dianisidine staining

o-dianisidine staining was performed as described previously ¹⁸. In brief, embryos were dechorionated, anesthetized and incubated in fresh staining solution (0.6 mg/mL o-dianisidine, 0.01 M sodium acetate pH4.5, 0.65% H₂O₂ and 40% Ethanol) for 20 minutes in dark. Washed three

times with methanol, and performed benzyl alcohol/benzyl benzoate (BABB) tissue clearing before imaging with stereo microscope (Leica, 160F).

Whole-mount *in situ* hybridization (WISH)

Whole-mount *in situ* RNA hybridization was performed as described previously ¹¹⁷. Antisense probes were synthesized using a digoxigenin RNA labeling kit (Roche, 11277073910). Primer sequences for all WISH probes used in this paper are provided in Table S4.

Quantitative real-time RT-PCR

Total RNA was isolated from embryos using Trizol (Invitrogen) and cDNA was generated using HiScript® III RT SuperMix (Vazyme) according to the manufacturer's instruction. Quantitative real-time PCR was performed using Lightcycler (Roche) and ChamQ SYBR qPCR Master Mix (Vazyme). Primer sequences are listed in Supplementary Materials Table S5. Gene expression level was normalized against GAPDH level.

Light-sheet fluorescence microscopy imaging

Light-sheet microscopy imaging was performed as described previously ¹¹⁸. In brief, transgenic zebrafish embryos were collected and maintained at 28.5°C. At the required developmental stages, the embryos were carefully dechorionated, paralyzed with tricaine, and transferred into 1% ultrapure low melting point agarose (16520-050; Invitrogen). The embryo was then drawn into a glass tube in top-down position using a 1 mL syringe with an 18G blunt needle. After agarose coagulation, a wire was inserted from bottom to push the agarose with embryo upwards, removed excess agar until the head area is exposed from the top of the glass tube. Finally, the glass tube was fixed in a sample holder for subsequent imaging.

Imaging was carried out with Luxendo Multi-View Selective-Plane Illumination Microscopy (Bruker). Optical calibrations were adjusted according to the instruction manual.

The optical plates are emitted from two Nikon CFI Plan Fluor 10x W 0.3 NA immersion objectives in opposite directions. The detection is completed by two Olympus 20x 1.0 NA immersion lenses. The parameters are set as follows: the green fluorescence channel uses a laser of 488 nm coupled with BP497-554 filter, the red fluorescent channel uses laser 561 nm coupled with BP580-627 filter, with an exposure time of 100 ms, a delay of 11 ms, a line mode of 50 px, and z-Stack interval of 3 µm. The raw data were stored in H5 file format, processed into TIFF format and merged Multi-View dataset using MATLAB software. After pre-processing, multi-fluorescence channels were merged in ImageJ, and 3D visualization and measurement were performed by Imaris.

Single-cell RNA sequencing

For knockout samples, 300 ng/µL Cas9 mRNA and a mixture of number 1 to 4 *betaPix* gRNAs with 50 ng/µL each were co-injected into wildtype embryos at one-cell stage. Injection with equivalent concentration of Cas9 mRNA with PBS served as siblings. After microinjection, embryos were collected and maintained at 28.5°C. At 24 or 48 hpf, zebrafish were dechorionated, paralyzed and transferred onto agarose plate. Heads were harvested with dissecting scissors in cold sterile 1X PBS with pooling 200 heads for each group. Heads were then dissociated in 900 µL Accutase cell detachment solution (Sigma-Aldrich, A6964) at 28.5°C for 3 hours and re-suspended by pipetting every 30 min. Once digestion was complete, 100 µL FBS was added to cell suspension and centrifuged at 4°C, 500 g for 3 min. The cells were gently re-suspended in cold 500 µL 2.5% FBS in 1X PBS and filtered through a 40 µm strainer. PI and Hoechst33342 were stained for distinguishing living cells. The single, living cells were sorted by Aria SORP (BD biosciences) into 1.5 ml tubes. The cell counts and vitality were verified by AOPI staining coupled with an Automated Cell Counter (Countstar BioTech). Single cells were barcoded with Chromium Next GEM Single Cell 3 'Reagent Kits V3.1 kit (10x Genomics, 1000269) in 10× Chromium Controller (10× Genomics). After qualified

by peak shapes, fragment size and tailing with Fragment Analyzer System kit (Agilent Technologies, DNF-915), single-cell transcriptome libraries were sequenced via Illumina High Throughput Sequencing PE150 (Novogene, Beijing). The sequencing data were analyzed using the CellRanger-6.1.1 (10x Genomics) and mapped to reference genome GRCz11-GRCz11.103.

Cells were excluded from subsequent analyses under the following conditions: when the number of expressed genes was less than 500, when there were abnormally high counts of UMIs or genes (outliers of a normal distribution), or when the mitochondrial content exceeded 9%. Unsupervised clustering was performed using Seurat (version: 4.0.2) with a resolution of 2.5, resulted in 71 cell populations and further annotated into 24 zebrafish major cranial cell types. Differential expression analysis in Seurat v4 was used to identify cluster/cell type markers by Wilcoxon rank sum test.

Scratch assay

U251 cells were planted into culture plate and performed transfection at the desired density. At 24 hours post-transfection, we vertically scratched monolayer cells by using a 200 μ L pipette tip. Washed the cells three times with PBS, then replaced with serum-free culture medium for further culture. Stereo fluorescence microscopy (Leica, 160F) was used to document the scratch size at 0 hours, 18 and 24 hours. Images were processed by ImageJ.

Immunostaining

For assessing tubulin expression, U251 cells were planted into chamber slides (Saining, 1093000) and transfected at the desired density. At 24 hours post-transfection, washed the cells once on ice with pre-cooled PBS, then fixed with pre-cooled methanol at -20°C for 20 minutes. The fixed cells were permeabilized with pre-cooled acetone at -20°C for 1 minute. Removed acetone, incubated 0.5% BSA/PBS blocking solution at room temperature for 15 minutes. Cells were then incubated with 1:200 anti- α -tubulin mouse monoclonal antibody (EASYBIO, BE0031) at 37°C for 45 minutes. After rinsing once with PBS, cells were incubated with 1:400 Alexa Fluor™ 488 goat anti-mouse IgG (H+L) secondary antibody (Invitrogen, A11029) at 37°C for another 45 minutes. Washed three times with PBS, and sealed with Mounting Medium with DAPI (ZSGB Bio, ZLI-9557). Images were acquired by upright fluorescence microscopy (Leica, DM5000B) at $40\times$ magnification, and processed by ImageJ.

Statistical analysis

Statistical analysis was performed using Graphpad Prism 6. The statistical significance of differences between the two groups was determined by the independent unpaired Student's t-test. Among three or more groups, one-way ANOVA analysis coupled with Dunnett's test were used. All data are presented as the mean $\pm$ SEM. P value <0.05 indicates significant, with individual P values mentioned in the figure/figure legends.

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Reviewer #1 (Public review):

The manuscript by Chiu et al describes the modification of the Zwitch strategy to efficiently generate conditional knockouts of zebrafish betapix. They leverage this system to identify a surprising glia-exclusive function of betapix in mediating vascular integrity and angiogenesis. Betapix has been previously associated with vascular integrity and angiogenesis in zebrafish, and betapix function in glia has also been proposed. However, this study identifies glial betapix in vascular stability and angiogenesis for the first time.

The study derives its strength from the modified CRISPR-based Zwitch approach to identify the specific role of glial betapix (and not neuronal, mural, or endothelial). Using RNA-in situ

hybridization and analysis of scRNA-Seq data, they also identify delayed maturation of neurons and glia and implicate a reduction in stathmin levels in the glial knockouts in mediating vascular homeostasis and angiogenesis. The study also implicates a betapix-zfhx3/4-vegfa axis in mediating cerebral angiogenesis.

There is both technical (the generation of conditional KOs) and knowledge-related (the exclusive role of glial betapix in vascular stability/angiogenesis) novelty in this work that is going to benefit the community significantly.

While the text is well written, it often elides details of experiments and relies on implicit understanding on the part of the reader. Similarly, the figure legends are laconic and often fail to provide all the relevant details.

Specific comments:

(1) While the evidence from cKO's implicating glial betapix in vascular stability/angiogenesis is exciting, glia-specific rescue of betapix in the global KOs/mutants (like those performed for stathmin) would be necessary to make a water-tight case for glial betapix.

(2) Splice variants of betapix have been shown to have differential roles in haemorrhaging (Liu, 2007). What are the major glial isoforms, and are there specific splice variants in the glial that contribute to the phenotypes described?

(3) Liu et al, 2012 demonstrated reduced proliferation of endothelial cells in bbh fish and linked it to deficits in angiogenesis. Are there proliferation/survival defects in endothelial cells in the glial KOs?

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Reviewer #2 (Public review):

Summary:

Using a genetic model of beta-pix conditional trap, the authors are able to regulate the spatio-temporal depletion of beta-pix, a gene with an established role in maintaining vascular integrity (shown elsewhere). This study provides strong in vivo evidence that glial beta-pix is essential to the development of the blood-brain barrier and maintaining vascular integrity. Using genetic and biochemical approaches, the authors show that PAK1 and Stathmins are in the same signaling axis as beta-pix, and act downstream to it, potentially regulating cytoskeletal remodeling and controlling glial migration. How exactly the glial-specific (beta-pix driven-) signaling influences angiogenesis or vascular integrity is not clear.

Strengths:

(1) Developing a conditional gene-trap genetic model which allows for tracking knockin reporter driven by endogenous promoter, plus allowing for knocking down genes. This genetic model enabled the authors to address the relevant scientific questions they were interested in, i.e., a) track expression of beta-pix gene, b) deletion of beta-pix gene in a cell-specific manner.

(2) The study reveals the glial-specific role of beta-pix, which was unknown earlier. This opens up avenues for further research. (For instance, how do such (multiple) cell-specific signaling converge onto endothelial cells which build the central artery and maintain the blood-brain barriers?)

Weaknesses:

Major:

(1) The study clearly establishes a role of beta-pix in glial cells, which regulates the length of the central artery and keeps the hemorrhages under control. Nevertheless, it is not clear how this is accomplished.

a. Is this phenotype (hemorrhage) a result of the direct interaction of glial cells and the adjacent endothelial cells? If direct, is the communication established through junctions or through secreted molecules?

b. The authors do not exclude the possibility that the effects observed on endothelial cells (quantified as length of central artery) could be secondary to the phenotype observed with deletion of glial beta-pix. For instance, can glial beta-pix regulate angiogenic factors secreted by peri-vascular cells, which consequently regulate the length of the central artery or vascular integrity?

c. The pictorial summary of the findings (Figure 7) does not include *Zfhx* or *Vegfa*. The data do not provide clarity on how these molecules contribute (directly or indirectly) to endothelial cell integrity. *Vegfa* is expressed in the central artery, but the expression of the receptor in these endothelial cells is not shown. Similarly, all other experimental analyses for *Zfhx* and *Vegfa* expression were performed in glial cells. More experimental evidence is necessary to show the regulation of angiogenesis (of endothelial cells) by glial beta-pix. Is the *Vegfa* receptor present on central arteries, and how does glial depletion of beta-pix affect its expression or response of central artery endothelial cells (both pertaining to angiogenesis and vascular integrity).

(2) Microtubule stabilization via glial beta-pix, claimed in Figure 5M, is unclear. Magnified images for h-beta-pix OE and h-stmn-1 glial cells are absent. Is this migration regulated by beta-pix through its GEF activity for *Cdc42/Rac*?

(3) Hemorrhages are caused by compromised vascular integrity, which was not measured (either qualitatively or quantitatively) throughout the manuscript. The authors do measure the length of the central artery in several gene deletion models (2I, 3C, 5F/J, 6G/K), which is indicative of artery growth/ angiogenesis. How (if at all) defects in angiogenesis are an indication of hemorrhage should be explained or established. Do these angiogenic growth defects translate into junctional defects at later developmental timepoints? Formation and maintenance of endothelial cell junctions within the hemorrhaging arteries should be assessed in fish with deleted beta-pix from astrocytes.

(4) More information is required about the quality control steps for 10X sequencing (Figure 4, number of cells, reads, etc.). What steps were taken to validate the data quality? The EC groups, 1 and 2-days post-KO are not visible in 4C. One appreciates that the progenitor group is affected the most 2 days post-KO. But since the effects are expected to be on the endothelial cell group as well (which is shown in in vivo data), an extensive analysis should be done on the EC group (like markers for junctional integrity, angiogenesis, mesenchymal interaction, etc.). Are Stathmins limited to glial cells? Are there indicators for angiogenic responses in endothelial cells?

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